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# Nonparametric validation of similar distributions and assessment of goodness of fit

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**Summary.** In this paper the problem of assessing the similarity of two cumulative distribution functions $F$ and $G$ is considered. An asymptotic test based on an $\alpha$-trimmed version of Mallows distance $\Gamma_\alpha(F, G)$ between $F$ and $G$ is suggested, thus demonstrating the similarity of $F$ and $G$ within a preassigned $\Gamma_\alpha(F, G)$ neighbourhood at a controlled type I error rate. The test proposed is applied to the validation of goodness of fit and for the nonparametric assessment of bioequivalence. It is shown that $\Gamma_\alpha(F, G)$ can be interpreted as average and population equivalence. Our approach is illustrated by various examples.

*Keywords*: Clinically relevant difference; Equivalence testing; Mallows distance; Model validation; Population equivalence; $p$-value curve; Testing goodness of fit

## 1. Introduction

One of the main goals of statistical inference is the assessment of a significant difference between several populations. Rejection of the hypothesis of equality at a controlled error rate leads to the empirically founded knowledge of a difference between populations. However, if the hypothesis cannot be rejected, often a large $p$-value is considered as sufficient evidence for the validity of the null hypothesis. The following example illustrates the above procedure, which is sometimes called the power approach.

### *1.1. Example 1*

In a multicentre clinical study performed at the Biometrical Centre, Department of Medical Statistics at Göttingen University, the cholesterol and fibrinogen levels were measured from 116 and 141 patients in two different clinical centres. Fig. 1 displays the corresponding histograms and estimators of the densities.

It was of interest to investigate whether the choice of centre has any effect on the result. A preliminary Kolmogorov–Smirnov (KS) goodness-of-fit test indicates that the data do not follow a normal distribution. From Table 1 it can be seen that the $p$-values (in parentheses) are smaller than 0.1 in each group and centre. The values in front of the $p$-values are the outcomes of the two-sided KS statistic. A two-sample Wilcoxon–Mann–Whitney (WMW)

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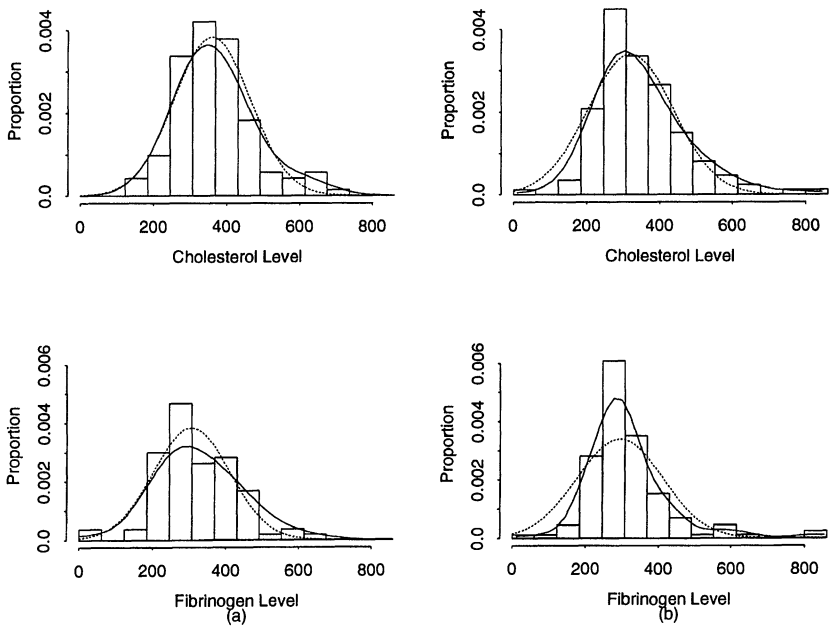


**Fig. 1.** Histograms with density estimates (———) and normal approximations (·····) for the multicentre clinical trial data in example 1: (a) centre 1; (b) centre 2

**Table 1.** KS test of composite normality for the multicentre clinical study

| *Centre* | *KS statistics for the following tests:* | |
|---|---|---|
| | *Cholesterol level* | *Fibrinogen level* |
| 1 | 0.095 (0.012) | 0.092 (0.066) |
| 2 | 0.136 (0.000) | 0.161 (0.000) |

test for the cholesterol and fibrinogen levels was performed. This leads to a $p$-value of 0.02 in the first case and to 0.33 in the second case, i.e. we know at a controlled error rate of $\alpha = 0.05$ that there is a difference between the clinical centres with respect to the cholesterol levels. The rather large $p$-value for the fibrinogen levels indicates homogeneous distributions at both centres. In comparison, a standard two-sample $t$-test gave a $p$-value of 0.11 for the cholesterol levels and 0.58 for the fibrinogen levels. In accordance with common practice one would omit the clinical centre as an influential factor for the fibrinogen levels within a (non)parametric linear model.

The following two questions arise. Firstly, how large is the difference within cholesterol levels, and, secondly, how large is the evidence for the homogeneity within fibrinogen levels?

The first question arises from the unsatisfactory observation that the pure test decision of rejecting the null hypothesis $F = G$ alone does not contain any information about the level of difference (Staudte and Sheather, 1990; Victor, 1987). Therefore, it is only of interest to show a scientifically relevant difference in contrast with a purely statistical difference. The second problem has recently been well documented in applications (see Rogers *et al.* (1993) and MacKinnon (1992) for a discussion in psychology and econometrics). From a methodo-

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logical point of view this question is directly related to bioequivalence testing which has become a challenging field during the last two decades (Mandallaz and Mau, 1981; Schuirmann, 1987; Chow and Liu, 1992). In broad terms, the 'equivalence way' is to replace the classical null hypothesis of equality by a suitable 'neighbourhood' which becomes the alternative. Most popular in bioequivalence testing is the assessment of the similarity of the means of certain pharmacokinetic parameters, to claim the same therapeutic effect of different formulations of a drug. Various researchers have criticized this concept (Anderson and Hauck, 1990; Holder and Hsuan, 1993) because *average bioequivalence* focuses only on a comparison of the means of the underlying distributions. Therefore, Hauck and Anderson (1992) suggested a bioequivalence criterion which requires the *entire distribution* of the test formulation to be sufficiently similar to the reference formulation. For a patient starting on a new drug, it was concluded that this criterion, which is known as population equivalence, seems to be more appropriate than average bioequivalence.

The aim of this paper is to provide a test for the assessment of similarity of distributions within a purely nonparametric framework. In Section 4, we show that this test can be used as a nonparametric test for population as well as average equivalence. In particular, the flexible choice of trimming allows robust bioequivalence assessment which is required in many practical situations (see Chow and Tse (1990)). Crucial for the success of a nonparametric equivalence test is the choice of the measure of discrepancy between probability distributions. Although it is necessary that this measure should be not too complicated it is also important that it contains all the relevant information for the experimenter. In this paper, we propose the use of a trimmed version of the $p$th Mallows (1972) distance

$$\Gamma_{\alpha,p}(F, G) := (1-2\alpha)^{-1}\left\{\int_{\alpha}^{1-\alpha} |F^{-1}(u) - G^{-1}(u)|^p \, \mathrm{d}u\right\}^{1/p} \tag{1}$$

between two distributions $F$ and $G$ in

$$\mathcal{F}_p := \left\{F\colon\ F \text{ is a cumulative density function (CDF) and } \int |x|^p \, \mathrm{d}F(x) < \infty\right\}.$$

Here $\alpha \in [0, \frac{1}{2})$ denotes a trimming bound and $p \geqslant 1$. If $p = 2$ we say Mallows distance and write $\Gamma_\alpha$. If in addition $\alpha = 0$, we write $\Gamma$. Confidence intervals for $\Gamma_\alpha$ and asymptotic tests for the testing problem

$$H\colon \Gamma_\alpha(F, G) > \Delta_0 \qquad \textit{versus} \qquad K\colon \Gamma_\alpha(F, G) \leqslant \Delta_0 \tag{2}$$

and its dual problem $K$ *versus* $H$ will be provided in the next section. As a more precise measure of the evidence of the test decision we propose the corresponding asymptotic $p$-value function given a fixed outcome of observations. This is illustrated in two examples in Section 4. In contrast with the test for the classical null $\Gamma(F, G) = 0$ (de Wet and Venter, 1972), for equivalence testing additional estimation of the variance of the empirical Mallows distance is required, which is numerically rather cumbersome. Therefore in Section 3 we report briefly on the finite sample behaviour of the proposed tests from a comprehensive simulation study. In particular, recommendations governed by sample size for the choice of trimming are given.

A numerical power comparison with the (asymptotic optimal) standard test for bioequivalence under normality shows that Mallows equivalence is even slightly more powerful in many cases.

Finally, the data of example 1 are reanalysed and a significant difference of the cholesterol levels between the clinical centres as well as for the fibrinogen levels is demonstrated.

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Section 2.1 is pertinent to those with an interest in applications exclusively. Sections 3 and 4 are understandable without previous reading of the more technical Section 2.2.

## 2. Asymptotic theory of the Mallows distance

### 2.1. Properties of Mallows distance

Throughout this paper, we assume $F$ and $G$ to be continuous CDFs.

$$F_n(x) = n^{-1} \sum_{i=1}^{n} \mathbb{I}_{(-\infty,x]}(X_i)$$

denotes the empirical distribution function of an independent and identically distributed (IID) sample $X_1, \ldots, X_n \sim F$ and

$$F_n^{-1}(t) = \inf\{x\colon F_n(x) \geqslant t\} = \sum_{i=1}^{n} X_{(i)}\, \mathbb{I}_{\{(i-1)/n<t\leqslant i/n\}} - \infty\, \mathbb{I}_{\{t=0\}}, \tag{3}$$

its left continuous inverse. Here, $X_{(i)}$ is the $i$th order statistic of a random sample of size $n$. The following result shows that a small (trimmed) Mallows distance $\Gamma_{\alpha,p}$ between $F$ and $G$ implies that all (trimmed) moments up to the $p$th moment are close, as well as the entire (trimmed) CDFs. We shall see in Section 4 that this property corresponds exactly to what has been demanded by various researchers for a nonparametric measure of bioequivalence.

*Lemma 1* (Dobrushin, 1970; Mallows, 1972; Bickel and Freedman, 1981). Let $F_n, F \in \mathcal{F}_p$, $p \geqslant 1$. Then $(\mathcal{F}_p, \Gamma_p)$ is a complete metric space. The following statements are equivalent:

(a) $\Gamma_p(F_n, F) \to 0$ as $n \to \infty$;

(b) $F_n \xrightarrow{\mathrm{d}} F$ and $\int |x|^p \,\mathrm{d}F_n(x) \to \int |x|^p \,\mathrm{d}F(x)$ as $n \to \infty$.

The next result shows that the Mallows distance controls the (trimmed) difference in means. The proof is an application of Jensen's inequality. In the following, we write $\gamma_\alpha$ for $\Gamma_\alpha^2(F, G)$ and $\gamma$ for $\Gamma^2(F, G)$.

*Lemma 2*. For $X \sim F$ and $Y \sim G$ with $F, G \in \mathcal{F}_{\alpha,2}$ the following inequalities are valid:

(a) $\gamma_\alpha^{1/2} \geqslant \Gamma_1^\alpha(F, G) \geqslant \dfrac{1}{1-2\alpha}\left|\displaystyle\int_{F^{-1}(\alpha)}^{F^{-1}(1-\alpha)} x\, F(\mathrm{d}x) - \int_{G^{-1}(\alpha)}^{G^{-1}(1-\alpha)} x\, G(\mathrm{d}x)\right|$,

(b) $\gamma^{1/2} \geqslant \Gamma_1(F, G) \geqslant |E(X-Y)|$.

Inequality (a) will fail to hold when the scaling in expression (1) is replaced in the definition of the trimmed Mallows distance by $(1-2\alpha)^{-1/p}$ as could be done as well. Further, we have in general $\Gamma_{\alpha,p} \geqslant (1-2\alpha)^{p(p-1)}\Gamma_{\alpha,p-1}$. For the special case of location–scale families $F = H\{(x-\eta)/\tau\}$ and $G = H\{(x-\mu)/\sigma\}$, we obtain for $H \in \mathcal{F}_2$

$$\gamma = (\mu-\eta)^2 + EX^2(\sigma-\tau)^2 + 2(\sigma-\tau)(\mu-\eta)EX, \tag{4}$$

where $X \sim H$. In the case $EX^2 = 1$ and $EX = 0$, the Mallows distance reduces to the Euclidean distance $\gamma^{1/2} = \{(\mu-\eta)^2 + (\sigma-\tau)^2\}^{1/2}$ defined in the parameter space $\mathbb{R} \times \mathbb{R}_+$. In the presence of trimming, we obtain for symmetric location–scale families

$$\gamma_\alpha^{1/2} = (1-2\alpha)^{-1/2}\left[(\mu-\eta)^2 + (\sigma-\tau)^2\left\{1 - 2\,\frac{z_{1-\alpha}\, h(z_{1-\alpha})}{1-2\alpha}\right\}\right]^{1/2},$$

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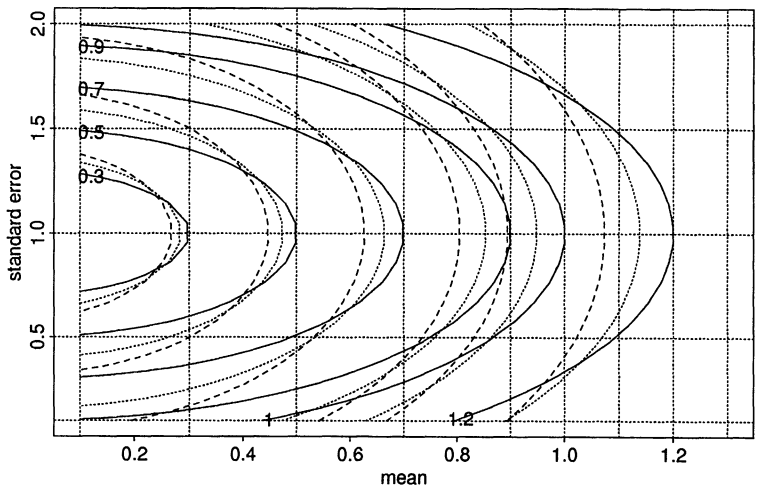


**Fig. 2.** Mallows distance contours ($F \sim N(0, 1)$ and $G \sim N(\mu, \sigma)$) for no trimming (———), 10% trimming (·····) and 20% trimming (- - - -)

where $h$ denotes the density of $H$ and $z_\alpha$ denotes the $\alpha$-quantile of $H$, i.e. $H(z_\alpha) = \alpha$. Trimming has the effect of reducing the squared distance (after rescaling by $1 - 2\alpha$) by $2z_{1-\alpha}\, h(z_{1-\alpha})(\sigma - \tau)^2(1 - 2\alpha)^{-1}$. If $h$ is a normal density (see also Fig. 2), this correction is roughly $2/3(\sigma - \tau)^2$ when $\alpha = 0.05$, i.e. trimming puts slightly more weight on the mean difference than on the difference of the scales. In the particular case of homogeneity of the variances, we have $\gamma_\alpha^{1/2} = |\mu - \eta|/(1 - 2\alpha)^{1/2}$.

## 2.2. Asymptotic theory

The following results give the asymptotic distribution of the estimated Mallows distance in the one- and two-sample case. From representation (12) in the proof of the following theorem given in Appendix B, it is apparent that the quantile process $q_n^F(z) := n^{1/2}\{F_n^{-1}(z) - F^{-1}(z)\}$ determines the asymptotic behaviour of $\Gamma_\alpha(F_n, F)$. A well-developed theory for the quantile process exists (see Shorack (1972), Czörgö and Revesz (1981) and Czörgö and Horvath (1993)) from which we obtain the regularity conditions on $F$ required to guarantee weak convergence to a weighted Brownian bridge

$$q_n^F(\cdot) \overset{D}{\Rightarrow} \frac{\mathbb{B}_0(\cdot)}{F' \circ F^{-1}(\cdot)}.$$

Here, $\mathbb{B}_0(\cdot)$ denotes a Brownian bridge defined on the space of cadlag functions ($D[0, 1]$, $||(\cdot)||_\infty$) equipped with the sup-norm, i.e. a centred Gaussian process with covariance function $st - s \wedge t$ (see for example Pollard (1984) for a careful description of $D$) and $s \wedge t$ and $s \vee t$ denote respectively the minimum and maximum of $s$ and $t$. To deal with the untrimmed case $\alpha = 0$ we must consider sequences of trimming bounds $(\alpha_n)_{n \in \mathbb{N}}$. The following regularity conditions for the sequence of trimming bounds and on the CDFs $F$ and $G$ are required to guarantee asymptotically a normal law for $n^{1/2}\{\Gamma_{\alpha_n}^2(F_n, G) - \gamma_\alpha\}$.

(a) Let $F \in \mathcal{F}_2$ be a continuous CDF $F$ with continuously differentiable density $F' = f > 0$. There is a real constant $\delta$ such that

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$$\sup_{x\in\mathbb{R}}\left[F(x)\{1-F(x)\}\left|\frac{f'(x)}{f^2(x)}\right|\right] \leqslant \delta.$$

(b) Let $\frac{1}{2} > (\alpha_n)_{n\in\mathbb{N}} > 0$ denote a sequence of trimming bounds, such that $\lim_{n\to\infty}(\alpha_n) \downarrow \alpha \geqslant 0$ and $\alpha_n \geqslant C\log\{\log(n)\}/n$ holds for a positive real constant $C > 0$.

(c) Assume that

$$\int_{\alpha_n}^{1-\alpha_n} \frac{1}{f\circ F^{-1}(u)}\,\mathrm{d}u = o\{n^{1/2}/\log(n)\}.$$

(d) Assume that

$$\gamma_{\alpha_n}^{1/2} - \gamma_\alpha^{1/2} = o(n^{-1/2}).$$

The following asymptotic result in the one-sample case will be used for the validation of goodness of fit in the sense of expression (2) where $G$ is a specific CDF. For example, proving goodness of fit to a standard normal distribution is required if we are interested in verifying the model assumptions of a (generalized) linear model. For this, note that the Studentized residuals for large samples are approximately a sample from a standard normal distribution if the model assumptions are correct (see for example Cook and Weisberg (1986), p. 56). We return to this problem in example 2, in Section 4.

*Theorem 1.* Let $\gamma_\alpha < \infty$ and assume conditions (a)–(d). Then it follows that

$$n^{1/2}\{\Gamma^2_{\alpha_n}(F_n, G) - \gamma_\alpha\} \overset{D}{\Rightarrow} \frac{2}{(1-2\alpha)^2}\int_0^{1-\alpha}\left\{\int_{\alpha\vee s}^{1-\alpha} h_{(F,G)}(t)\,\mathrm{d}t\right\}\mathrm{d}\mathbb{B}(s),$$

where $\mathbb{B}(s)$ denotes a Brownian bridge in $(D[0, 1], \|\cdot\|_\infty)$ and

$$h_{(F|G)}(t) := \frac{F^{-1}(t) - G^{-1}(t)}{f\circ F^{-1}(t)}.$$

Further, if

$$\int_{\alpha\vee s}^{1-\alpha} h_{(F|G)}(t)\,\mathrm{d}t \in L_2(\lambda), \tag{5}$$

then the distribution of $n^{1/2}\{\Gamma^2_{\alpha_n}(F_n, G) - \gamma_\alpha\}$ is asymptotically normal with mean 0 and variance

$$\sigma^2_\alpha(F|G) := \frac{4}{(1-2\alpha)^4}\left(\int_0^{1-\alpha}\left\{\int_{\alpha\vee s}^{1-\alpha} h_{(F|G)}(t)\,\mathrm{d}t\right\}^2\mathrm{d}s - \left[\int_0^{1-\alpha}\left\{\int_{\alpha\vee s}^{1-\alpha} h_{(F|G)}(t)\,\mathrm{d}t\right\}\mathrm{d}s\right]^2\right).$$

The next theorem provides the asymptotic theory in the two-sample case, which will be used to assess the nonparametric equivalence of two distributions.

*Theorem 2.* Let $(X_i)_{i=1,\ldots,m}$ $((Y_i)_{i=1,\ldots,n})$ be IID according to $F \in \mathcal{F}_2$ $(G \in \mathcal{F}_2)$ and assume that $(\alpha_{(n\wedge m)})_{n,m\in\mathbb{N}}$ is a sequence of trimming bounds such that asumption (b) holds. Assume further that conditions (a), (c) and (d) hold for $F$ and $G$. Let $m\wedge n \to \infty$ such that $n/(n+m) \to \lambda \in (0, 1)$. If $\gamma_\alpha < \infty$, then it follows that

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$$\left(\frac{nm}{n+m}\right)^{1/2}\{\Gamma^2_{\alpha_n\wedge m}(F_m, G_n)-\gamma_\alpha\} \overset{\mathrm{D}}{\Rightarrow} \frac{2}{(1-2\alpha)^2}\left[\sqrt{\lambda}\int_0^{1-\alpha}\left\{\int_{\alpha\vee s}^{1-\alpha} h_{(F|G)}(t)\,\mathrm{d}t\right\}\mathrm{d}\mathbb{B}_1(s)\right.$$
$$\left.+\sqrt{(1-\lambda)}\int_0^{1-\alpha}\left\{\int_{\alpha\vee s}^{1-\alpha} h_{(G|F)}(t)\,\mathrm{d}t\right\}\mathrm{d}\mathbb{B}_2(s)\right],$$

where $\mathbb{B}_1(\cdot)$ and $\mathbb{B}_2(\cdot)$ denote two independent Brownian bridges on $(D[0, 1], \|\cdot\|_\infty)$. Further, if

$$\int_{\alpha\vee s}^{1-\alpha} h_{(G|F)}(t)\,\mathrm{d}t,\ \int_{\alpha\vee s}^{1-\alpha} h_{(F|G)}(t)\,\mathrm{d}t \in L_2(\lambda) \tag{6}$$

holds, then the distribution of $\{nm/(n+m)\}^{1/2}\{\Gamma^2_{\alpha_n\wedge m}(F_m, G_n)-\gamma_\alpha\}$ is asymptotically normal with mean 0 and variance

$$\sigma^2_\alpha(F, G) := \lambda\,\sigma^2_\alpha(F|G) + (1-\lambda)\,\sigma^2_\alpha(G|F).$$

For the proof see Appendix B.

*Remark 1.* At first glance, assumptions (a)–(d) seem to be rather restrictive. However, the choice of positive trimming $\alpha > 0$ guarantees a bounded support and the only requirement that we need is $f > 0$. In this case assumption (d) is guaranteed by assumption (b). We still require the more general condition (d) to deal with the case $\alpha = 0$. In a location–scale model, it follows from equation (4) that assumption (d) is always satisfied. In general $\alpha_n = o(n^{-1/2})$ is sufficient for condition (d) to hold. Conversely, condition (b) serves as a lower bound and we obtain that $\alpha_n = n^{-\delta}$, $\frac{1}{2} < \delta < 1$ is always sufficient for theorem 1 to hold. In practice, however, the trimming bound $\alpha$ can always be chosen to be rather small, but it must be taken into account that the asymptotic distribution will be a rather inaccurate approximation for the finite sample distribution of the empirical Mallows distance. In this case, conditions (b) and (d) indicate that the trimmed statistic should not be replaced by the untrimmed statistic too soon. With this argument assumptions (a)–(d) are not real restrictions for the distributions to guarantee the limit law; rather, they are indicators for those cases where the asymptotic convergence might deteriorate. This will be the case when the distributions have heavy tails (condition (c)) and when the density is strongly peaked or almost zero (condition (a)). These observations are supported by a simulation study which will be presented in the next section.

In the case $\sigma_\alpha(F, G) = 0$ we have

$$\{nm/(n+m)\}^{1/2}\{\Gamma^2_{\alpha_n\wedge m}(F_m, G_n)-\gamma_\alpha\} \xrightarrow{\mathrm{p}} 0 \qquad \text{as } n\wedge m \to \infty \tag{7}$$

which happens in particular for $F = G$. This is explained by the fact that in this case $\gamma_\alpha$ is a von Mises functional of first-order degeneracy. A non-degenerate weak limit is obtained when expression (7) is multiplied by $\{nm/(n+m)\}^{1/2}$. The resulting distribution was calculated by de Wet and Venter (1972) for the case of $F = G$ standard normal. They used this result to obtain a goodness-of-fit test for testing the hypothesis $H$: $F = \Phi$. However, our aim is to provide tests for hypotheses (2) and the converse problem where $\Delta_0 > 0$ is a positive bound. The proof of the following theorem is straightforward by observing that the convergence in expression (7) is sufficient to prove consistency of our tests. Let $u_\alpha$ denote the $\alpha$-quantile of the standard normal distribution and $\hat{\sigma}^2_\alpha(F, G)$ be any consistent estimator of the variance $\sigma^2_\alpha(F, G)$. For brevity, in what follows we consider the two-sample case only; the one-sample case is analogous by theorem 1.

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*Theorem 3.* Let $(X_i)_{i=1,\dots,m}$ $((Y_i)_{i=1,\dots,n})$ be IID according to $F \in \mathcal{F}_2$ $(G \in \mathcal{F}_2)$ and let $\hat{\sigma}_\alpha(F_m, G_n)$ be a consistent estimator of $\sigma_\alpha$. Under the assumptions of theorem 2, the test with critical region

$$\left(\frac{nm}{n+m}\right)^{1/2} \frac{\Gamma^2_{\alpha_n \wedge m}(F_m, G_n) - \Delta_0^2}{\hat{\sigma}_\alpha(F_m, G_n)} \leqslant u_\alpha \tag{8}$$

is a consistent test for the equivalence problem $H$ against $K$ in expression (2) and

$$\left(\frac{nm}{n+m}\right)^{1/2} \frac{\Gamma^2_{\alpha_n \wedge m}(F_m, G_n) - \Delta_0^2}{\hat{\sigma}_\alpha(F_m, G_n)} \geqslant u_{1-\alpha} \tag{9}$$

is a consistent test for the difference problem $K$ against $H$.

*Remark 2.* Theorem 3 includes the consistency in the degenerate case $K$: $\gamma_\alpha = 0$. In particular, we find that consistency remains valid even for sequences of hypotheses $H_{n,m}$: $\gamma_\alpha^{1/2} > \Delta_{0,m,n}$ as long as $\Delta_{0,m,n}^{-1} = o\{(n \vee m)^{1/2}\}$. This implies that $nm/(n+m)\,\Gamma^2_\alpha(F_m, G_n) < \infty$ almost surely, i.e. we can still assess asymptotically with probability 1 the similarity of $F$ and $G$ as long as the sequence of hypotheses $H_{n,m}$ approaches $K$: $\gamma_\alpha = 0$ with a slower rate than $n^{-1/2}$.

*Remark 3.* From theorem 3 an asymptotic one-sided $1-\alpha$ confidence interval for $\gamma_\alpha$ is obtained as $[0, \Gamma^2_{\alpha_n \wedge m}(F_m, G_n) + u_{1-\alpha}\,\hat{\sigma}_\alpha(F_m, G_n)(1/n + 1/m)^{1/2}]$. We shall see, however, in the next section, that this may become a rather crude approximation if $\gamma_\alpha$ is rather small.

## 3. Estimators for $\Gamma$ and $\sigma^2$ and the small sample properties of the equivalence test: simulation results

To perform the test given in theorem 3, we must estimate $\Gamma_\alpha$ and the variance $\sigma^2_\alpha(F, G)$. When trimming is present these estimators are rather complicated and can be found in Appendix A. The corresponding S-PLUS functions are available by request from the authors where all estimators are computed by replacing the CDFs and their inverses by their empirical counterparts and evaluating $\Gamma_\alpha(F_m, G_n)$ and $\sigma^2_\alpha(F_m, G_n)$ as a Riemannian sum. In the untrimmed case ($\alpha = 0$) this simplifies to

$$\Gamma^2(F_n, G_n) = \frac{1}{n}\sum_{i=1}^{n} (y_{(i)} - x_{(i)})^2$$

for equal sample sizes, and to

$$\Gamma^2(F_m, G_n) = \frac{1}{m}\sum_{i=1}^{m} X^2_{(i)} + \frac{1}{n}\sum_{i=1}^{n} Y^2_{(i)} - 2\sum_{i=1}^{m}\sum_{i=1}^{n} X_{(i)} Y_{(j)} \lambda_{ij}$$

for unequal sample sizes $n \neq m$, where

$$\lambda_{ij} = \left(\frac{i}{m} \wedge \frac{j}{n} - \frac{i-1}{m} \vee \frac{j-1}{n}\right) \mathbb{I}_{\{i/m \wedge j/n > (i-1)/m \vee (j-1)/n\}}$$

and $X_{(0)} := X_{(1)}$ and $Y_{(0)} := Y_{(1)}$. To investigate the small sample properties of the Mallows equivalence test we performed detailed Monte Carlo studies which are presented comprehensively in Czado and Munk (1996). We briefly summarize the basic results which address the following questions.

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(a) How large must samples sizes $n$ and $m$ be for the Mallows test to maintain its nominal level and to achieve reasonable power?
(b) What is a reasonable choice of the trimming bound $\alpha$?
(c) How does the Mallows test perform under non-symmetric, heavy-tailed and strongly peaked distributions for $F$ and $G$ (see remark 1)?

For the first three simulation studies $F$ (or $G$) was assumed standard normal (normal with mean $\mu$ and standard error $\sigma$) and a sample of size $m$ (or $n$) was drawn from $F$ (or $G$). Values for $\mu$ ranged between 0.1 and 2.6 and for $\sigma$ between 0.5 and 1.5. In the first study, we investigated large equal sample sizes ($n = m = 50$, 100, 200), whereas in the second study we focused on small equal sample sizes ($n = m = 10$, 15, 20, 25). For the third study, unequal sample sizes were studied ($n = 100$ and $m = 30$, 50, 75, 90). For large samples trimming constants of $\alpha = 0.05$ and $\alpha = 0.1$ were investigated; for small samples, $\alpha = 0$ and $\alpha = 0.1$. 500 data sets from each parameter setting were generated for the first two studies and 250 data sets were generated for the third study, and the Mallows equivalence test of $H$: $\Gamma_\alpha(F, G) \geqslant \Delta_0$ *versus* $K$: $\Gamma_\alpha(F, G) < \Delta_0$ for several $\Delta_0$-values between 0.3 and 1.2 were performed at significance level $\alpha_s = 0.05$.

The results from these studies can be summarized as follows. The Mallows equivalence test is found to be always liberal. This liberality decreases as the sample size and/or tolerance bound $\Delta_0$ is increased. In particular, if the tolerance bound $\Delta_0 > 1$, then sample sizes of $n$, $m \geqslant 25$, as long as $\frac{1}{2} \leqslant m/n \leqslant 2$ is satisfied, are sufficient for a maximal deviation of 0.03 between the nominal and the actual level. If the tolerance bound $\Delta_0 \ll 1$, then larger sample sizes of $n$, $m > 50$ are necessary. In particular, the equivalence test becomes always liberal rather than conservative. For the appropriate difference test the converse effect was observed. This is to be expected since the asymptotic law for the Mallows distance fails for $\gamma_\alpha(F, G) = 0$. If we can assume homogeneous variances then a slight trimming of 5–10% is preferable rather than no trimming, since it increases the power of the Mallows test. This shows trimming as a powerful way to make robust against outliers. Nevertheless, as homogeneity of the variances increases, trimming leads to a more liberal test.

To address question (c) the Mallows equivalence test was investigated under a family of generalized logistic distributions $\{F_\psi, \psi \in \mathbb{R}\}$ (for a definition see Czado (1992)) with heavier right tail ($\psi < 1$) and lighter right tail ($\psi < 1$) than the logistic distribution ($\psi = 1$). For $\psi \gg 1$, the corresponding density is highly peaked. Here, $n = 25$, 50 and 100 with $m/n = 0.96$ and $m/n = 0.48$ were investigated. It was observed that in the heavy tail case ($\psi < 1$) the nominal size was better maintained for 16% trimming than for 8% trimming. However, the observed power is adequate for $n \geqslant 25$. In contrast, for the highly peaked case ($\psi > 1$), the nominal size was maintained for both trimmings, but the observed power was inadequate for $n = 25$ and $n = 50$. Therefore, the conclusions in remark 2 were exactly supported. To summarize, the effect of heavy-tailed and strongly peaked densities decreases the quality of the approximation. It was found that as in the normal case the bound $\Delta_0 = 1$ serves as a changepoint for the quality of the asymptotic normal approximation to the finite sample distribution.

## 4. Applications and examples

### *4.1. Nonparametric bioequivalence testing*

In bioequivalence studies, when normality of the data is assumed, the two one-sided test procedures should be applied in a $2 \times 2$ crossover design as recommended by various

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regulatory authorities (Food and Drug Administration, 1992; European Commission, 1993). If this distributional assumption fails a distribution-free rank sum (WMW) test has been suggested by various researchers (see Hauschke *et al.* (1990) or Hsu *et al.* (1994)). This test, however, should be applied only when a pure location difference between test and reference formulation can be assumed because otherwise the test decision may be extremely misleading. As pointed out by Hauck and Anderson (1992) it is important to compare the entire distributions which should include specifically the between-subject variability of bioavailability (see Chow and Liu (1992), pages 186 and 218) to guarantee exchangeability in the test formulations. When an equivalence test is based on two independent samples lemmas 1 and 2 guarantee that these requirements are exactly fulfilled by the Mallows distance $\Gamma_\alpha(F, G)$ because the first two trimmed moments are close whenever $\Gamma_\alpha$ is small. Hence, the proposed test applies to population and average equivalence, simultaneously. One might argue that the Mallows test cannot be applied to both periods in a two- (or more) period crossover design, as it is standard when carry-over effects can be excluded. This is not caused, however, by the particular choice of the measure of equivalence $\Gamma_\alpha(F, G)$. It is merely that *no* nonparametric measure allows for a proper analysis in a crossover design because the interpretation of the test decision can be distorted heavily by the dependence structure of the observations. Only within a semiparametric linear model (as required by the WMW test) does a crossover design remain tractable.

Another common problem encountered with decision-making in bioequivalence studies consists of the detection of outlying observations because these have dramatic effects on the bioequivalence tests (Chow and Tse, 1990; Liu and Weng, 1991). Therefore, a trimmed version of the Mallows statistic may be especially useful to the applied working statistician. In this case, we find from the discussion in Section 2 that testing bioequivalence by using the trimmed Mallows statistic is tantamount to the assessment of similar trimmed moments and distributions, simultaneously.

A key issue in the nonparametric bioequivalence discussion is the sample size required to control a small probability of type II error. Therefore Czado and Munk (1996) simulated the relative efficiency of the Mallows equivalence test and the standard equivalence test under a linear model with additive normal errors. Typically, such a model occurs after a logarithmic transformation of the raw data (such as the area under the blood concentration–time curve or the time to achieve maximal concentration) where the ratio of the means must be within some reasonable limits. When we assume a multiplicative linear model the parameter of interest is $|\mu_T - \mu_R|$, where $\mu_T$ and $\mu_R$ denote the mean of the test and reference drug respectively (Mandallaz and Mau, 1981). The trimmed Mallows distance reduces after rescaling exactly to this quantity, i.e.

$$\gamma_\alpha^{1/2}(1-2\alpha)^{1/2} = |\mu_T - \mu_R|.$$

Surprisingly, the simulation study showed that the power of the standard test was in most cases slightly exceeded by the Mallows test, although the standard test can be shown to be asymptotically optimal. This curious finding can be explained by the fact that only when the unknown variance $\sigma^2$ is rather small compared with the equivalence bound $\Delta_0$ does the asymptotic optimality allow for a valid interpretation in realistic sample sizes, say $m, n \approx 20$. Otherwise, the power tends uniformly to 0 as pointed out by Brown *et al.* (1995). In addition, the Mallows test was found always to be liberal, which causes an additional improvement in power.

It is interesting that trimming does not affect the finite sample approximation under the

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standard model because variance homogeneity in both groups is assumed. Therefore, the Mallows test represents a valid and powerful tool in bioequivalence assessment to guard against outliers.

### 4.2. *p*-value curves associated with Mallows test

To obtain a more precise insight into the evidence against the test decisions in inequalities (8) and (9), the additional consideration of the asymptotic $p$-value function corresponding to the hypothesis $H$ and $K$ in expression (2) is helpful. In the one-sample case $X \sim F$ and $G$ fixed, this asymptotic $p$-value function is given by

$$P(\Delta_0) = \lim_{n\to\infty} \sup_{\{F: F\in H\}} [P_F\{T(X) > t\}] = 1 - \Phi(t) \tag{10}$$

if the sample $X = x$ is observed. Here $T(X)$ denotes the Mallows test statistic $n^{1/2}\{\Gamma^2_{\alpha_n}(F_n, G) - \Delta_0^2\}/\sigma_\alpha(F_n, G)$ and $t$ the corresponding observed statistics at $X = x$. The two-sample case is similar. The function $P(\Delta_0)$ can be interpreted as follows. If $P(\Delta_0) \leqslant \alpha$, the data lead to the rejection of the hypothesis of non-equivalence $H$: $\Gamma(F, G) \geqslant \Delta_0$ at level $\alpha$, whereas by symmetry we reject the hypothesis of equivalence $H$: $\Gamma(F, G) \leqslant \Delta_0$ whenever $P(\Delta_0) \geqslant 1 - \alpha$. For a careful discussion of the use of $p$-values corresponding to precise interval hypotheses as a measure of evidence in the parametric case, see Berger and Delampady (1987) and the subsequent discussion. Although they gave strong reasons against a naïve use of $p$-values, we agree with Cox's rejoinder (p. 335). Sometimes a substantial improvement on $p$-values may be possible; however, the conclusion that $p$-values have no role is incorrect. In particular, equation (10) shows that $P(\cdot)$ equals asymptotically the $p$-value function of a one-sided test. Pratt (1965) showed that this allows for a valid Bayesian interpretation in a parametric set-up as a measure of evidence for $H$.

### 4.3. Example 1 (continued)

Alternatively, the nonparametric tests given in Section 2 were applied to this problem. Fig. 3 gives the corresponding $p$-value curves for 1.4%, 4.3% and 10% trimming for fibrinogen and cholesterol levels. From the curves for 1.4% trimming, we see that an absolute difference of 15 and of 8 at level $\alpha = 0.1$ are present for the fibrinogen and cholesterol levels respectively. When we standardize these values, this corresponds to about 5% and 2% of the medians of the fibrinogen and cholesterol levels respectively. For the fibrinogen levels, this significantly contradicts the finding of the WMW test, which can be explained by the fact that this test focuses only on $\int F\,\mathrm{d}G$ as a measure of discrepancy and not on the entire distributions. From these curves it is further concluded that the cholesterol levels between the two centres are equivalent, if we tolerate a difference $\Delta_0 = 42$ which corresponds to about 12% of the medians for the two samples. We can show a similar equivalence for the fibrinogen levels if at least a 4.3% trimming is allowed. For 1.4% trimming equivalence with regard to 15% of the medians can still be shown. This indicates outliers in the data, which is also evident from Fig. 1.

To summarize, the example illustrates a systematic error whenever assessing equality of distributions by means of accepting the null hypothesis (power approach). Although we could show similar situations with the Mallows equivalence test for the cholesterol and the fibrinogen levels it is found to be different when using the WMW test, because the underlying test criterion focuses on a misleading measure of discrepancy $\int F\,\mathrm{d}G$ and, hence, is unable to detect a significant difference in one situation, although it is equally evident in the other situation. The next example illustrates the opposite effect. The KS test behaves too sensitively

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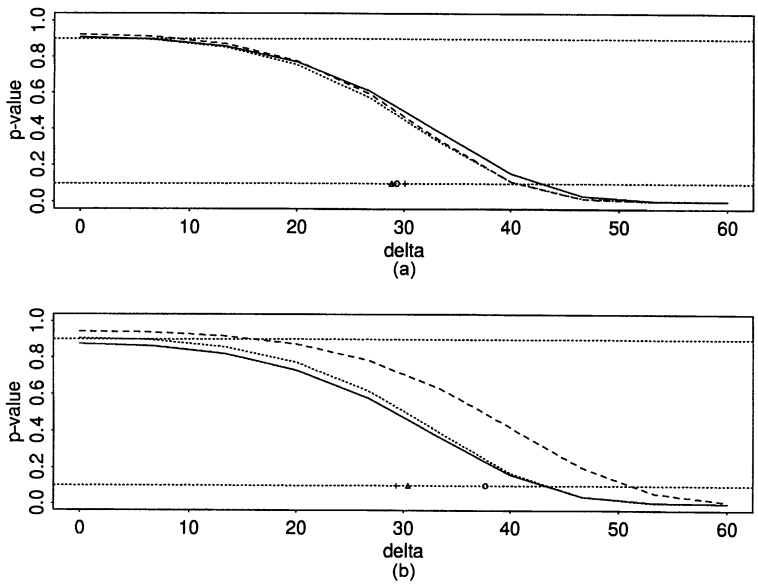


**Fig. 3.** Two-sample $p$-value curves for the multicentre clinical data: (a) cholesterol (———, 9.92% trimming; ·····, 4.26% trimming; - - - -, 1.42% trimming); (b) fibrinogen (———, 10.08% trimming; ·····, 4.32% trimming; - - - -, 1.44% trimming)

and detects a difference which is not larger than in a situation where no difference is indicated.

### 4.4. Example 2

Steinhoff *et al.* (1995) investigated the influence of epilepsy on extroceptive suppression of temporalis (es1) and masseter (es2) muscle activity. They compared the extroceptive suppression of muscle activity of 31 epileptic patients and 20 normal controls. Measurements were taken from each subject at the left and right temporalis and masseter. Further, measurements for each subject have been averaged to form the variables es1 and es2. The results of an explanatory analysis are given in Table 2.

This indicates that only the es2-values for patients show evidence of non-normality. To investigate goodness of fit to a normal distribution we standardize the data and use the one-sample Mallows equivalence test. Note that this involves the estimation of the mean and variance by its sample estimators. The asymptotic results of Section 2 are therefore not immediately applicable, and extensions of results for proving composite normality are currently considered. However, for illustration, the corresponding one-sample $p$-value curves are plotted in Fig. 4 for the two variables of the patients group. It shows equivalence at $\Delta_0 = 0.4$ and $\Delta_0 = 0.5$ for the standardized es1- and es2-values respectively. This corresponds to $0.4 \times 3.67 \approx 1.5$ and $0.5 \times 11.71 \approx 5.9$ on the original scale and therefore to approximately 8% and approximately 13% of the means for es1 and es2 respectively. If we allow for 19.4% trimming for the es2-data, only then can a significant difference of 0.1 on the standardized scale be detected. A closer look at the data shows a cluster of three observations in the far right-hand tail. Removing these shows a more non-normal histogram, explaining the last finding. To summarize, the one-sample nonparametric equivalence test agrees with

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**Table 2.** KS test of composite normality, estimated mean and standard error for example 2

| | | *Values for the following parameters:* | |
|---|---|---|---|
| | | *es1* | *es2* |
| Normal | KS statistic | 0.1431 (0.351) | 0.1569 (0.221) |
| | Mean | 19.75 | 42.94 |
| | Standard error | 3.72 | 9.26 |
| Patients | KS statistic | 0.1081 (0.460) | 0.1769 (0.0146) |
| | Mean | 19.24 | 42.45 |
| | Standard error | 3.67 | 11.71 |

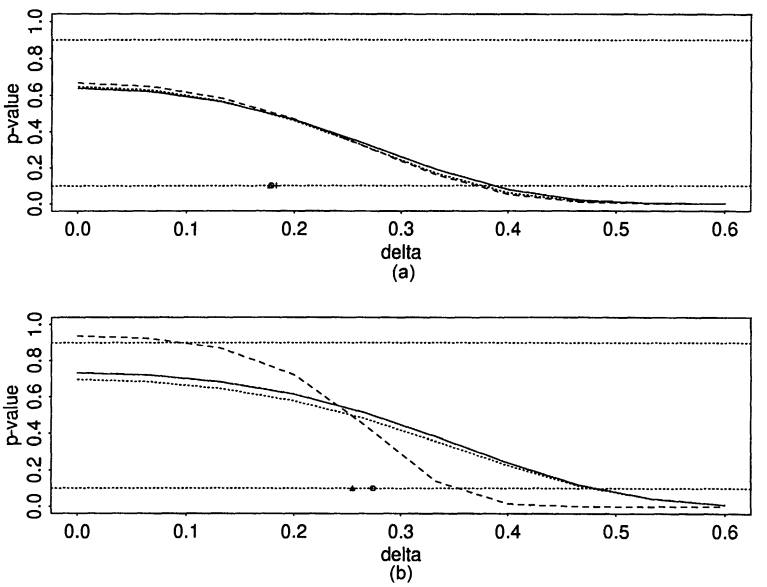


**Fig. 4.** One-sample $p$-value curves for the variables es1 and es2 for the patients group: (a) es1 (———, 6.45% trimming; ·····, 12.9% trimming; - - - -, 19.35% trimming); (b) es2 (———, 6.45% trimming; ·····, 12.9% trimming; - - - -, 19.35% trimming)

the KS test for composite normality for the variable es1. For variable es2, however, there is not much more evidence to assume non-normality (as the KS test indicates falsely). Finally, the $p$-value curves for the two-sample nonparametric equivalence test are given in Fig. 5 for the normal subjects and patients. They prove similarity of es1-values for the normal subjects and patients up to $\Delta_0 = 1.2$ which corresponds to $100 \times 1.2/19.5\% \approx 6\%$ of the mean. The similarity of the es2-values for the two groups only occurs at $\Delta_0 = 6.2$ which corresponds to $100 \times 6.2/42.94\% \approx 15\%$. Ultimately, the clinician must decide whether these values should be regarded as empirical evidence for the similarity or for a scientifically significant difference of the underlying distributions.

## 5. Conclusions and discussion

It has been illustrated by two examples that the assessment of similarity of two distributions

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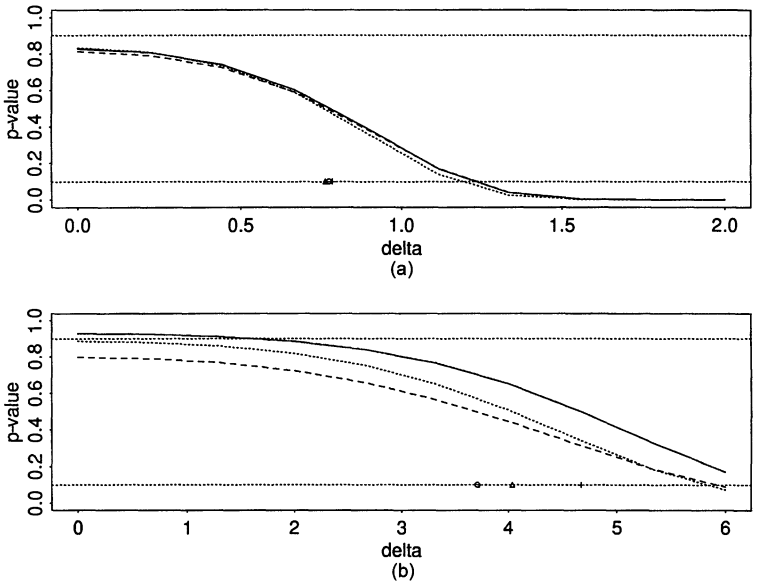


**Fig. 5.** Two-sample $p$-value curves for the es1- and es2-data: (a) es1 (———, 6.46% trimming; ·····, 12.9% trimming; - - - -, 19.36% trimming); (b) es2 (———, 6.46% trimming; ·····, 12.9% trimming; - - - -, 19.36% trimming)

from accepting the hypothesis of equality may lead to the wrong decision with high probability, even when large $p$-values are observed. But even when we reject the null hypothesis of equality the pure test decision gives no information about a scientifically significant difference between the distributions.

To overcome these difficulties, a nonparametric measure of discrepancy between two distributions which generalizes the Mallows distance was proposed. On the basis of this measure, tests for precise interval hypotheses were provided. Moreover, the associated $p$-value function represents a very informative tool for the exact level of similarity as well as for a significant difference. We conclude that when testing goodness of fit the classical power approach to 'prove the null hypothesis' by accepting equality at a rather large level $\alpha$ is insufficient and should additionally be supported by a consideration of the Mallows $p$-value function. Our approach applies also to the nonparametric assessment of bioequivalence because the proposed measure of discrepancy has an immediate interpretation of the direct drug effect in terms of average as well as population bioequivalence. In addition to the between-subject variability similarity of the entire distributions can be tested in contrast with the commonly applied WMW test. Nevertheless, one problem occurs which is not known from testing the classical null hypothesis of equality. In most applications the main difficulty will consist in the determination of precise bounds for which the distributions are regarded as similar or scientifically different.

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## Appendix A (estimators for $\gamma_\alpha$ and $\sigma_\alpha^2$)

For $\alpha = a/m$, $a$ an integer between 1 and $n$, and $m \geqslant n$ the Mallows distance in the two-sample case is estimated as follows:

$$\Gamma_\alpha(F_m, G_n) = \frac{1}{1-2\alpha}\left(\frac{1}{m}\sum_{i=a+1,[(i-1)n/m]=[in/m]}^{m-a}(x_{(i)} - y_{([in/m])})^2 + \sum_{i=a+1,[(i-1)n/m]=[in/m]-1}^{m-a}\left[(x_{(i)} - y_{([(i-1)n/m])})^2 \times \left\{\frac{[(i-1)n/m]}{n} - \frac{i-1}{m}\right\} + (x_{(i)} - y_{([in/m])})^2\left\{\frac{i}{m} - \frac{[(i-1)n/m]}{n}\right\}\right]\right)^{1/2}$$

where $[x]$ is the smallest integer greater than or equal to $x$. It remains to provide a consistent estimate $\hat{\sigma}_\alpha^2(F, G)$ of $\sigma_\alpha^2(F, G)$. For this, rewrite the asymptotic variance expression in theorem 1 as

$$\sigma_\alpha^2(F, G) = \frac{4}{(1-2\alpha)^4}[\lambda\{T_\alpha^2(F, G) - T_\alpha^1(F, G)^2\} + (1-\lambda)\{T_\alpha^4(F, G) - T_\alpha^3(F, G)^2\}].$$

Each term $T_\alpha^i(F, G)$ will now be estimated by $T_\alpha^i(F_m, G_n)$ for $i = 1, \ldots, 4$. For example, $T_\alpha^1(F, G)$ can be rewritten as

$$\begin{aligned} T_\alpha^1(F, G) &= \int_0^{1-\alpha}\int_{F^{-1}(\alpha\wedge s)}^{F^{-1}(1-\alpha)}[z - G^{-1}\{F(z)\}]\,\mathrm{d}z\,\mathrm{d}s \\ &= \tfrac{1}{2}\int_0^{1-\alpha}\{F^{-1}(1-\alpha)^2 - F^{-1}(\alpha\wedge s)^2\}\,\mathrm{d}s - \int_0^{1-\alpha}\int_{F^{-1}(\alpha\wedge s)}^{F^{-1}(1-\alpha)} G^{-1}\{F(z)\}\,\mathrm{d}z\,\mathrm{d}s. \end{aligned} \tag{11}$$

Assuming that $n = m$ and $\alpha = a/n$, $T_\alpha^1(F_n, G_n)$ can be calculated using equation (11) and substituting empirical CDFs $F_n$ and $G_n$ for $F$ and $G$ respectively as

$$\begin{aligned} T_\alpha^1(F_n, G_n) = \alpha&\left\{\tfrac{1}{2}(x_{(n-a)}^2 - x_{(a)}^2) - \sum_{i=a}^{n-a-1} y_{(i)}(x_{(i+1)} - x_{(i)})\right\} \\ &+ \frac{1-2\alpha}{2}x_{(n-a)}^2 - \frac{1}{2n}\sum_{i=a+1}^{n-a-1}x_{(i)}^2 - \frac{1}{n}\sum_{j=a+1}^{n-a-1}\sum_{i=j}^{n-a-1} y_{(i)}(x_{(i+1)} - x_{(i)}). \end{aligned}$$

Using the same approach for $T_\alpha^2(F_n, G_n)$ gives after some algebra

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$$T_\alpha^2(F_n, G_n) = \alpha\left\{\tfrac{1}{2}(x_{(n-a)}^2 - x_{(a)}^2) - \sum_{i=a}^{n-a-1} y_{(i)}(x_{(i+1)} - x_{(i)})\right\}^2 + \frac{1}{4n}\sum_{i=a+1}^{n-a-1}(x_{(n-a)}^2 - x_{(i)}^2)^2$$
$$-\frac{1}{n}\sum_{j=a+1}^{n-a-1}(x_{(n-a)}^2 - x_{(j)}^2)\sum_{i=j}^{n-a-1} y_{(i)}(x_{(i+1)} - x_{(i)}) + \frac{1}{n}\sum_{j=a+1}^{n-a-1}\sum_{i=j}^{n-a-1} y_{(i)}(x_{(i+1)} - x_{(i)}).$$

The expressions for $T_\alpha^3(F_n, G_n)$ and $T_\alpha^4(F_n, G_n)$ are the same as for $T_\alpha^1(F_n, G_n)$ and $T_\alpha^2(F_n, G_n)$ respectively except that the role of $x_{(i)}$ and $y_{(i)}$ is reversed. In the case of unequal sample sizes, boundary terms must be included. For $m \geqslant n$ we obtain

$$T_\alpha^1(F_m, G_n) = \alpha\left\{\tfrac{1}{2}(x_{(m-a)}^2 - x_{(a)}^2) - \sum_{i=a}^{m-a-1} y_{([in/m])}(x_{(i+1)} - x_{(i)})\right\} + \frac{1-2\alpha}{2}x_{(m-a)}^2$$
$$-\frac{1}{2m}\sum_{i=a+1}^{m-a} x_{(i)}^2 - \frac{1}{m}\sum_{j=a+1}^{m-a-1}\sum_{i=j}^{m-a-1} y_{([in/m])}(x_{(i+1)} - x_{(i)})$$

and

$$T_\alpha^2(F_m, G_n) = \alpha\left\{\tfrac{1}{2}(x_{(m-a)}^2 - x_{(a)}^2) - \sum_{i=a}^{m-a-1} y_{([in/m])}(x_{(i+1)} - x_{(i)})\right\}^2 + \frac{1}{4m}\sum_{i=a+1}^{m-a-1}(x_{(m-a)}^2 - x_{(i)}^2)^2$$
$$-\frac{1}{m}\sum_{j=a+1}^{m-a-1}(x_{(m-a)}^2 - x_{(j)}^2)\sum_{i=j}^{m-a-1} y_{([in/m])}(x_{(i+1)} - x_{(i)}) + \frac{1}{m}\sum_{j=a+1}^{m-a-1}\left\{\sum_{i=j}^{m-a-1} y_{([in/m])}(x_{(i+1)} - x_{(i)})\right\}^2.$$

The other terms $T_\alpha^i(F_m, G_n)$, $i = 3, 4$, can be expressed in a similar fashion.

## Appendix B (proofs)

For $\int f(u)\,\mathrm{d}u$ we shall simply write $\int f$ whenever the variable and domain of integration are obvious from the context.

### *B.1. Proof of theorem 1*

Condition (d) implies that

$$n^{1/2}\left\{\Gamma_{\alpha_n}^2(F_n, G) - \Gamma_{\alpha_n}^2(F, G)\right\} = n^{1/2}\frac{2}{(1-2\alpha_n)^2}\int_{\alpha_n}^{1-\alpha_n}(F_n^{-1} - F^{-1})(F^{-1} - G^{-1}) + n^{1/2}\Gamma_{\alpha_n}^2(F_n, F)$$
$$:= n^{1/2}(T_{1n} + T_{2n}) \tag{12}$$

and

$$\Gamma_{\alpha_n}^2(F, G) - \Gamma_\alpha^2(F, G) = o(n^{-1/2}).$$

If $f = F' > 0$, the corresponding quantile process

$$q_n^F(z) = n^{1/2}\{F_n^{-1}(z) - F^{-1}(z)\}$$

can be strongly approximated on compact sets by a sequence of weighted Brownian bridges $(\mathbb{B}_n(\cdot))_{n\in\mathbb{N}}$ in $D[0, 1]$ (see theorem 4.5.7 in Czörgö and Revesz (1981), p. 153) as follows:

$$\sup_{\alpha_n \leqslant y \leqslant 1-\alpha_n} |f \circ F^{-1}(y)\, q_n^F(y) - \mathbb{B}_n(y)| = O\{n^{-1/2}\log(n)\} \qquad \text{almost surely.} \tag{13}$$

We obtain

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$$\left| n^{1/2} \frac{(1-2\alpha_n)^2}{2} T_{1n} - \int_{\alpha_n}^{1-\alpha_n} \frac{\mathbb{B}_n(u)}{f \circ F^{-1}(u)} \{F^{-1}(u) - G^{-1}(u)\} \, \mathrm{d}u \right|$$

$$= \left| \int_{\alpha_n}^{1-\alpha_n} \{F^{-1}(u) - G^{-1}(u)\} \left\{ q_n^F(u) - \frac{\mathbb{B}_n(u)}{f \circ F^{-1}(u)} \right\} \mathrm{d}u \right|$$

$$\leqslant \sup_{\alpha_n \leqslant y \leqslant 1-\alpha_n} |q_n^F(y) f \circ F^{-1}(y) - \mathbb{B}_n(y)| \int_{\alpha_n}^{1-\alpha_n} \left| \frac{F^{-1}(u) - G^{-1}(u)}{f \circ F^{-1}(u)} \right| \mathrm{d}u$$

$$\leqslant \sup_{\alpha_n \leqslant y \leqslant 1-\alpha_n} |f \circ F^{-1}(y) \, q_n^F(y) - \mathbb{B}_n(y)| \, \Gamma_1^\alpha \, o\left\{ \frac{n^{1/2}}{\log(n)} \right\} = o(1) \qquad \text{almost surely}$$

where we have used equation (13), condition (c) and $\Gamma_1^\alpha \leqslant \Gamma^\alpha < \infty$ (see lemma 2). Taking into account that

$$\mathbb{B}_n(\cdot) \stackrel{\mathrm{D}}{=} \mathbb{B}_0(\cdot)$$

for each $n \in \mathbb{N}$, we find that the limit distributions of

$$\int_{\alpha_n}^{1-\alpha_n} \{F^{-1}(u) - G^{-1}(u)\} \, q_n^F(u) \, \mathrm{d}u$$

and

$$\int_{\alpha}^{1-\alpha} \{F^{-1}(u) - G^{-1}(u)\} \frac{\mathbb{B}_0(u)}{f \circ F^{-1}(u)} \, \mathrm{d}u$$

are equal. We have

$$n^{1/2} R_n \stackrel{\mathrm{D}}{\Rightarrow} Z = \int_{\alpha}^{1-\alpha} \frac{\mathbb{B}_0(s)}{f \circ F^{-1}(s)} \, \mathrm{d}s = \int_0^{1-\alpha} \int_{\alpha \vee s} \frac{1}{f \circ F^{-1}(t)} \, \mathrm{d}t \, \mathrm{d}\mathbb{B}_0(s),$$

where $R_n := n^{1/2} T_{2n}$. $Z$ is a random variate with finite variance and hence it follows that $R_n \to_{\mathrm{p}} 0$. This proves the first assertion. When condition (5) holds, straightforward calculation together with proposition 2.2.1 in Denker (1985) shows the second statement.

### *B.2. Proof of theorem 2*

The proof for the two-sample case follows the same pattern as the proof of theorem 1. Therefore we only sketch the argument. We must again replace $q_m^F$ and $q_n^G$ by two independent weighted Brownian bridges $\mathbb{B}_1(\cdot)/\{f \circ F^{-1}(\cdot)\}$ and $\mathbb{B}_2(\cdot)/\{g \circ G^{-1}(\cdot)\}$ and apply the estimates

$$\int \{(F_m^{-1} - G_n^{-1})^2 - (F^{-1} - G^{-1})^2\} = \int \{(F_m^{-1} - F^{-1})^2 + (G_n^{-1} - G^{-1})^2\} + \int \{2F_m^{-1}F^{-1} + 2G_n^{-1}G^{-1}$$
$$- 2(F^{-1})^2 - 2(G^{-1})^2 - 2F_m^{-1}G_n^{-1} + 2F^{-1}G^{-1}\}.$$

From

$$\left( \frac{nm}{n+m} \right)^{1/2} \int_{\alpha_n}^{1-\alpha_n} (F_m^{-1} - F^{-1})^2 \stackrel{\mathrm{p}}{\to} 0,$$

$$\left( \frac{nm}{n+m} \right)^{1/2} \int_{\alpha_n}^{1-\alpha_n} (G_n^{-1} - G^{-1})^2 \stackrel{\mathrm{p}}{\to} 0$$

it remains to estimate

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$$\int \{2F_m^{-1}F^{-1} + 2G_n^{-1}G^{-1} - 2(F^{-1})^2 - 2(G^{-1})^2 - 2F_m^{-1}G_n^{-1} + 2F^{-1}G^{-1}\}$$
$$= \int \{2F^{-1}(F_m^{-1} - F^{-1}) + 2G^{-1}(G_n^{-1} - G^{-1}) + 2F_m^{-1}(G^{-1} - G_n^{-1})\}$$
$$+ \int \{2G_n^{-1}(F^{-1} - F_m^{-1}) + 2(F^{-1} - F_m^{-1})(G^{-1} - G_m^{-1})\} \tag{14}$$

where the second integral of equation (14) asymptotically vanishes of order $o_p[\{nm/(n+m)\}^{-1/2}]$, which is a direct consequence of the Cauchy–Schwarz inequality. Now we apply theorem 1 and we obtain

$$\left(\frac{nm}{n+m}\right)^{1/2} \{\Gamma_{\alpha_n}^2(F_n, G_n) - \gamma_n\} \overset{\mathrm{D}}{\Rightarrow} \frac{2}{(1-2\alpha)^2} \left\{ \lambda \int_\alpha^{1-\alpha} \frac{F^{-1}(t) - G^{-1}(t)}{f \circ F^{-1}(t)} \mathbb{B}_1(t)\, \mathrm{d}t \right.$$
$$\left. - (1-\lambda) \int_\alpha^{1-\alpha} \frac{F^{-1}(s) - G^{-1}(s)}{g \circ G^{-1}(s)} \mathbb{B}_2(s)\, \mathrm{d}s \right\}.$$

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